$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a} + \hat{a}^+\right)$$

$$\hat{a}(\tau) = e^{-\omega\tau}\hat{a} \quad , \quad \hat{a}^+(\tau) = e^{\omega\tau}\hat{a}^+$$

$$G(\tau) = -\langle T_\tau\, \hat{x}(\tau)\,\hat{x}(0)\rangle = -\left(\frac{\hbar}{2m\omega}\right)\langle T_\tau \left(\hat{a}e^{-\omega\tau} + \hat{a}^+ e^{\omega\tau}\right)\left(\hat{a}+\hat{a}^+\right)\rangle$$

$$G(\tau) = -\theta(\tau)\left(e^{-\omega\tau}\langle \hat{a}\hat{a}^+\rangle + e^{\omega\tau}\langle \hat{a}^+\hat{a}\rangle\right)$$

$$-\theta(-\tau)\left(e^{\omega\tau}\langle \hat{a}\hat{a}^+\rangle + e^{-\omega\tau}\langle \hat{a}^+\hat{a}\rangle\right)$$

But $\langle \hat{a}^+\hat{a}\rangle = \langle \hat{n}\rangle = \dfrac{1}{e^{\beta\omega}-1} \underset{n_\omega}{\overset{\text{in}}{=}}$ thermal equilibrium

$$\Rightarrow \langle \hat{a}\hat{a}^+\rangle = 1 + \frac{1}{e^{\beta\omega}-1} = 1 + \langle \hat{n}\rangle$$

$$G(\tau) = -\left(\frac{\hbar}{2m\omega}\right)\Big[\theta(\tau)\left\{\left(1+\langle \hat{n}\rangle\right)e^{-\omega\tau} + \langle n\rangle e^{\omega\tau}\right\}$$

$$+\theta(-\tau)\left\{\langle \hat{n}\rangle e^{-\omega\tau} + \left(1+\langle n\rangle\right)e^{\omega\tau}\right\}\Big].$$

$$G(\omega_n) = \int_0^\beta d\tau\, G(\tau)\, e^{i\omega_n\tau} = -\frac{\hbar}{2m\omega}\int_0^\beta d\tau\, e^{i\omega_n\tau}\Big[(1+n)e^{-\omega\tau} + ne^{\omega\tau}\Big]$$

$$e^{i\omega_n\beta} = e^{i2\pi} = 1.$$

$$\int_0^\beta d\tau\, e^{\tau(i\omega_n \pm \omega)} = \left.\frac{e^{\tau(i\omega_n\pm\omega)}}{i\omega_n \pm \omega}\right|_0^\beta = \frac{e^{\pm\beta\omega}-1}{i\omega_n\pm\omega}$$

$$G(\omega_n) = \left(-\frac{\hbar}{2m\omega}\right)\left\{(1+n_\omega)\,\frac{e^{-\beta\omega}-1}{i\omega_n-\omega} + n_\omega\,\frac{e^{\beta\omega}-1}{i\omega_n+\omega}\right\}$$

but $e^{\beta\omega}-1 = \frac{1}{n_\omega} \Rightarrow e^{\beta\omega} = 1+\frac{1}{n_\omega} = \frac{n_\omega+1}{n_\omega}$

$$\Rightarrow e^{-\beta\omega}-1 = \frac{n_\omega}{n_\omega+1}-1 = -\frac{1}{n_\omega+1}$$

$$\Rightarrow G(\omega_n) = \left(-\frac{\hbar}{2m\omega}\right)\left\{\frac{-1}{i\omega_n-\omega} + \frac{1}{i\omega_n+\omega}\right\}$$